

UNIT 09

CHEMICAL KINETICS



Rate of Reaction

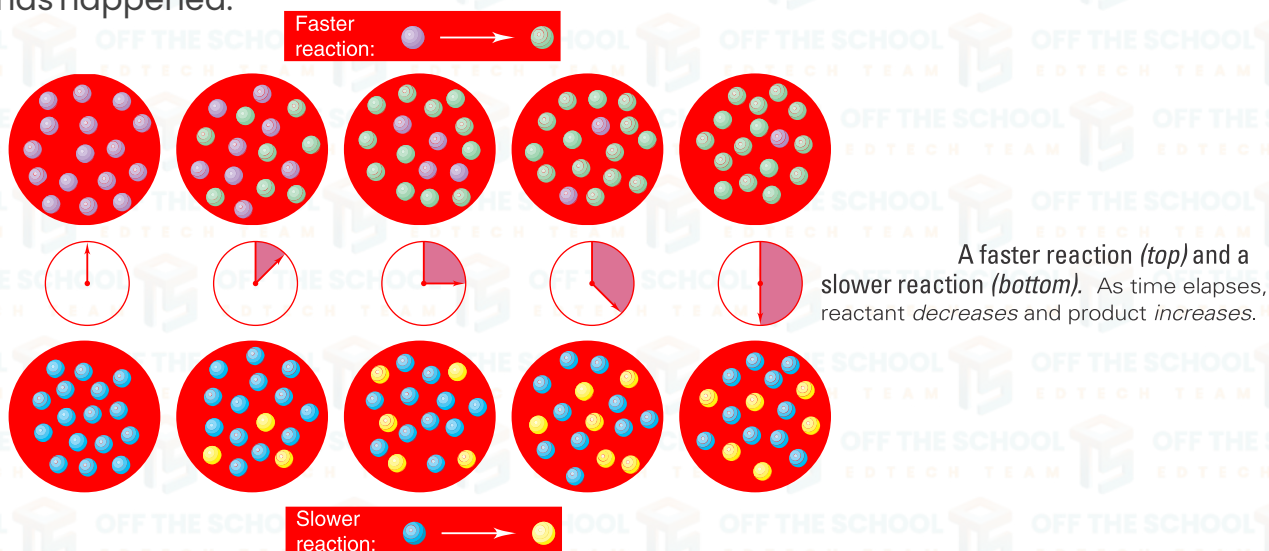
The rate of reaction is calculated using the change in concentration of reactants or products over a certain time period. The mathematical expression given for the rate of reaction is:

$$\text{Rate} = \Delta x / \Delta t$$

In this formula:

- Δx stands for the change in concentration of the reactant or product.
- Δt is the change in time over which this concentration change occurs.

The 'change in concentration' means how much the concentration goes up or down, and the 'change in time' means the amount of time over which this change has happened.



For a reaction where reactant A turns into product B, the rate at which A disappears is given by:

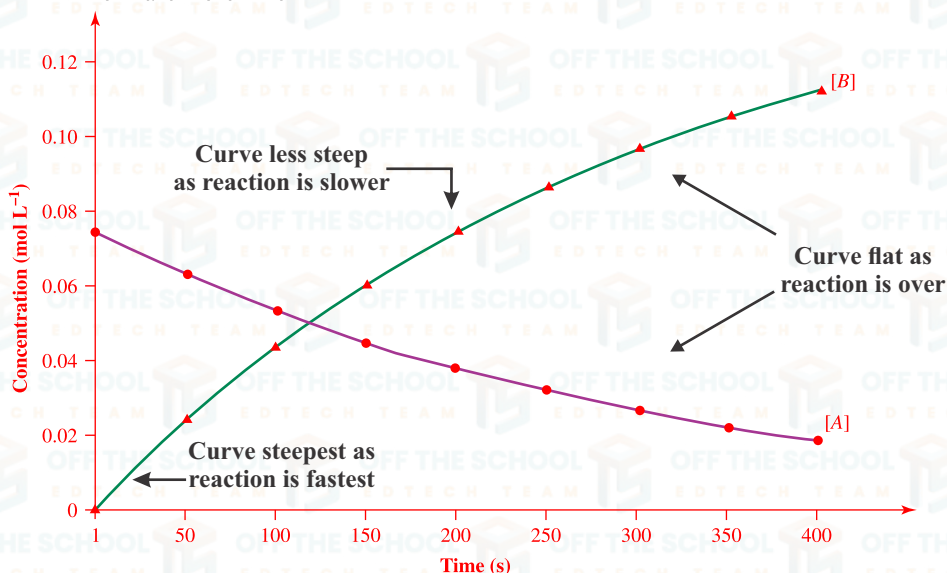
$$\text{Rate A disappearing} = -\frac{\Delta[A]}{\Delta t}$$

The negative sign indicates that the concentration of A decreases over time. And the rate at which B appears is given by:

$$\text{Rate B appearing} = \frac{\Delta[B]}{\Delta t}$$

Here, $[A]$ and $[B]$ represent the concentrations of A and B, respectively. If we're looking at a graph like the one in the picture, these rates correspond to the slopes of the lines for A and B on the graph: a steeper slope means a faster rate.

The progress (concentration–time profile) of the reaction $A \rightarrow 2B$, showing the concentrations of A and B over time



Rate law and rate expression:

The rate law for a general reaction where substance A reacts with substance B to produce a product can be written as:

$$\text{Rate} = k[A]^x[B]^y$$

In this expression:

- k is the rate constant, which changes with temperature but is independent of the concentration of the reactants.
- $[A]$ is the concentration of reactant A, and $[B]$ is the concentration of reactant B.
- x and y represent the reaction orders with respect to reactants A and B, respectively, and they describe how the rate of reaction depends on the concentration of each reactant. These are typically determined experimentally.

"The rate law expresses the rate of a chemical reaction in terms of the concentration of the reactants. The higher the concentration of reactants, the more molecules are available to collide and react, hence a faster rate of reaction. The rate law provides a quantitative relationship between the molar concentration of reactants and the rate of reaction."

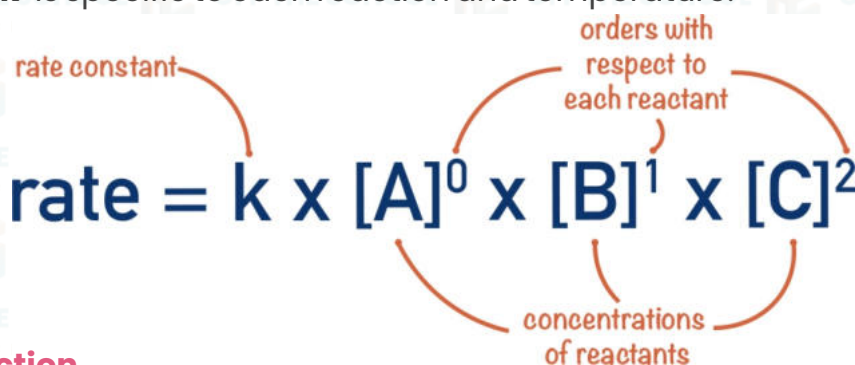
If the concentration of each reactant is unity (1 mol/dm^3), the rate law simplifies to the rate constant k , as the concentration terms raised to any power will be one:

$$\text{Rate} = k$$

The **specific rate constant** k at a given temperature can be defined as;

the rate of reaction when the molar concentration of each reactant is one.

The value of ' k ' is specific to each reaction and temperature.

**Order of Reaction**

Definition: The "order of reaction" refers to the power to which the concentration of a reactant is raised in the rate law. It provides information about the relationship between the concentration of the reactants and the rate of the reaction. The order is determined experimentally and is often a whole number, but it can also be a fraction or zero.

Here's how the order of reaction is related to the rate law:

- For a rate law given by $\text{Rate} = k[A]^x[B]^y$, the order with respect to each reactant is given by the exponents x and y .
- The overall order of the reaction is the sum of these exponents: $x + y$.



Units for Rate Constants:

- **Zero Order Reactions:** The rate is independent of the concentration of the reactants. The units for the rate constant (k) are moles per liter per second ($\text{mol/L}\cdot\text{s}$).
- **First Order Reactions:** The rate is directly proportional to the concentration of one reactant. The units for the rate constant are per second (s^{-1}).
- **Second Order Reactions:** The rate is proportional to the square of the concentration of one reactant or to the product of the concentrations of two reactants. The units for the rate constant are liters per mole per second ($\text{L/mol}\cdot\text{s}$).



shows reaction is zero order
with respect to concentration of A



shows reaction is first order
with respect to concentration of B



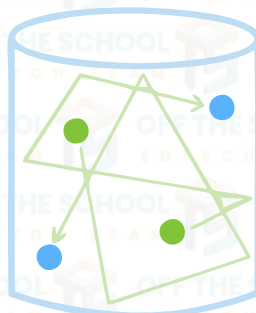
shows reaction is second order
with respect to concentration of C

Factors affecting rate of reaction:

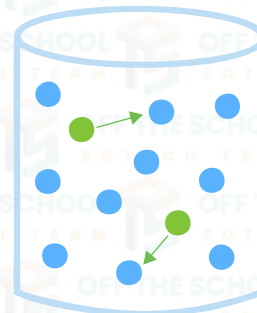
The factors affecting the rate of reaction are:

1. Nature of Reactants:

Reactants with different chemical structures and bonding types react at different rates. For example, sodium reacts quickly in air or water due to its high reactivity and tendency to lose its valence electron.



In this reaction, because the concentration of the reactants is small, it's hard for the green molecules to make collisions with the blue molecules



A higher concentration of blue molecules makes it more likely for the green molecules to make collisions, thus causing the reaction to move faster!

2. Concentration of Reactants:

An increased concentration of reactants leads to more collisions and a higher chance of reaction occurring.

The rate of reaction is therefore generally increased with higher reactant concentrations.



3. Surface Area:

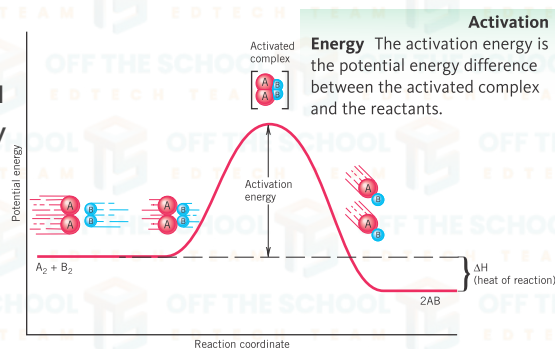
Finely divided materials or those with greater surface area react more quickly because there is more area for reactants to collide with, enhancing the rate of reaction.

4. Radiation (Light):

Photochemical reactions depend on the absorption of light. Light can provide the energy needed to initiate reactions, and increasing light intensity can speed up these reactions.

5. Temperature:

Raising the temperature generally increases the rate of a reaction. It's often observed that a 10°C increase in temperature can approximately double the reaction rate. This is due to an increase in the kinetic energy of molecules, leading to more frequent and energetic collisions that can overcome the activation energy barrier.



These factors are crucial in controlling the speed and outcome of chemical reactions for both industrial applications and laboratory research.

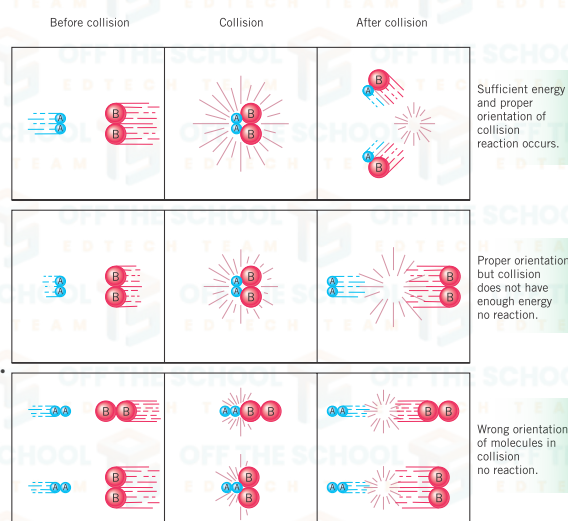
COLLISION THEORY:

This theory states that chemical reactions can only happen when reactant molecules crash into each other with enough energy and in the right orientation. The rate of reaction depends on how often and how effectively these collisions occur.

Activation Energy (E_a):

The extra amount of energy that reacting molecules must have for a reaction to occur.

It's the difference between the energy needed to start a reaction (threshold energy) and the average energy of the molecules. Activation energy can be seen as a barrier that molecules must overcome for a reaction to happen.



Reaction of A_2 and B_2 Reactions occur only when colliding molecules have the minimum amount of energy and the right orientation.

Expression:

$$E_a = \text{Threshold Energy} - \text{Average Internal Energy of Molecules}$$

Transition State:

A temporary, high-energy state where molecules have partially broken old bonds and are forming new ones. It's neither the reactants nor the products but an intermediate stage with a very short lifespan. Molecules in the transition state are called the "activated complex."

Catalysis:

It's a process where a substance, called a catalyst, speeds up a chemical reaction without being used up or changed permanently.

Catalyst:

A catalyst is like a helper that makes a chemical reaction go faster. It works by



providing an easier way for the reaction to happen, lowering the energy needed for the molecules to react. It's kind of like a shortcut that gets you to the same place but with less energy.

Characteristics of catalyst

- 1. Efficiency without High Temperature:** Catalysts allow chemical reactions to occur efficiently without the need for high temperatures.
- 2. Lower Activation Energy:** They provide a new pathway for the reaction with a lower activation energy barrier compared to the uncatalyzed pathway.
- 3. Involvement in Early Reaction Steps:** Catalysts are involved in the early stages of a reaction, forming an intermediate complex with the reactants.
- 4. Regeneration:** Catalysts are regenerated at the end of the reaction sequence; they emerge unchanged and can be reused.
- 5. Speed Up Reactions:** Positive catalysts increase the reaction rate by lowering the activation energy, leading to a greater number of effective collisions between molecules.
- 6. Inhibition:** Negative catalysts or inhibitors slow down the reaction rate.

Homogeneous catalysis:

Homogeneous catalyst is when the catalyst and the reactants are in the same phase (like both being liquids or gases), making them mix well and react smoothly.

It can be classified into two classes.

• Acid-Base Catalysis:

Acid-base catalysis involves the acceleration of a chemical reaction through the presence of acids or bases that donate or accept protons, respectively. The acid donates protons while the base accepts them, thereby altering the activation energy and facilitating the reaction.



• Autocatalysis:

Autocatalysis is a phenomenon where a reaction product acts as a catalyst for its own production. This leads to an increase in the rate of reaction over time as the concentration of the autocatalyst increases.



Heterogeneous Catalysis:

Heterogeneous catalysis happens when the catalyst is in a different phase than the reactants. Usually, this means the catalyst is a solid, and the reactants are either liquids or gases. In this process, reactions occur at the interface of the phases. An example equation might look like:



The reactant molecules adsorb onto the surface of the solid catalyst, react to form the products, and then desorb from the surface.

Enzyme Catalysis:

Enzyme catalysis is a specific type of biocatalysis where the catalyst is a biological molecule known as an enzyme. Enzymes are highly selective, working with specific reactants known as substrates. The general equation for enzyme catalysis is:



Enzymes bind to substrates, facilitate the chemical reaction to form products, and then release the products, remaining unchanged and ready to catalyze another reaction.

SIMPLE EXPLANATIONS FOR CHEMICAL BEHAVIORS:

(i) Powdered Marble Reacting with Hydrochloric Acid:

Powdered marble has a larger surface area compared to a solid lump, which means more marble is exposed to the acid at once. This allows the acid to react with more marble particles simultaneously, making the reaction faster.



(ii) Milk Souring Faster in Summer than Winter:

Warmer temperatures speed up chemical reactions, including the bacteria activity in milk that causes it to sour. In summer, it's warmer than in winter, so bacteria grow faster and make the milk sour quickly.

(iii) Reactants in Solution React Faster at High Concentration:

When reactants are more concentrated, there are more particles in the same space. This means they bump into each other more often, which can lead to more reactions happening in the same amount of time.

(iv) Increasing Temperature and Reaction Rate:

Raising the temperature gives particles more energy. When they have more energy, they move faster and collide more often with enough force to react, which increases the rate of reaction.

Difference between rate of reaction and rate constant:

Rate of Reaction	Rate Constant
Depends on concentration	Independent of concentration
Changes with temperature	Typically changes only with temperature
Units vary (mol/L·s, s ⁻¹ , L/mol·s)	Units depend on reaction order
Measured empirically	Calculated from rate and concentrations
Describes how quickly reactants are used up	A constant value specific to each reaction at a given temperature

Difference between homogenous and heterogenous catalyst:

Homogeneous Catalyst	Heterogeneous Catalyst
In the same phase as reactants	In a different phase than reactants
Forms a uniform mixture	Does not form a uniform mixture
Can be a liquid or gas	Usually, a solid
Reaction occurs in a single phase	Reaction occurs at the interface of two phases
Can offer more selectivity	May be less selective but more easily recovered



Differences: Positive Catalyst vs. Inhibitor

Positive Catalyst	Inhibitor
Speeds up a reaction	Slows down a reaction
Reduces the activation energy	Increases the activation energy or prevents the reaction
Not consumed in the reaction	May or may not be consumed
Used in small amounts	The amount used can vary
Does not change the direction of the reaction	Can prevent the reaction from proceeding

Differences: Elementary vs. Overall Reaction

Elementary Reaction	Overall Reaction
Consists of a single step	Consists of multiple steps
No intermediates are formed	May involve intermediates
Represents an individual reaction event	Represents the sum of all elementary steps in the mechanism
Cannot be broken down into simpler steps	Can be broken down into elementary steps
Rate law can be directly written from the reaction stoichiometry	Rate law is derived from the rate laws of the individual elementary steps

